

Overcoming shock instability of the HLL-type Riemann solvers



Liqi Liu^a, Xiao Li^a, Zhijun Shen^{a,b,*}

^a Institute of Applied Physics and Computational Mathematics, LCP, P. O. Box 8009-26, Beijing 100088, China

^b Center for Applied Physics and Technology, HEDPS, Peking University, China

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ABSTRACT

This paper concentrates on numerical shock instabilities of some approximate Riemann solvers which stem from the HLL method, including HLLC-type and HLLM-type flux functions. Carbuncle phenomena are illustrated for these numerical schemes, especially for those believed to be carbuncle-free, such as the HLL scheme. A linear matrix stability analysis shows that there exists a threshold independent of shock strength to trigger instability for a given numerical scheme. In this instability mechanism, a specific cell interface in the numerical structure of shock layer is regarded as the source of instability if the density ratio between both sides of this interface exceeds the threshold. The modification by adjusting nonlinear signal velocities in these numerical schemes can change solution behaviors in the shock layer, and reduce the degree of density transition on the specific interface. A very simple cure strategy for the HLL-type Riemann solvers is presented based on the in-depth understanding of the carbuncle onset of viscous shock.

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1. Introduction

The carbuncle phenomenon is a numerical instability that affects the capturing of shock waves when low dissipative, upwind schemes are used to numerically analyze high speed flows. It depends on many factors, such as mesh geometry, mesh size, flow Mach number, and specific heat ratio. In-depth understanding of the carbuncle onset contributes to design better algorithm to overcome difficulty in simulating strong shock wave problems.

Such phenomenon was first reported by Peery and Imlay [18] as they computed the supersonic flowfield around a blunt body with Roe's scheme. It also occurs in other situations such as the computation of a planar moving shock wave referred to as Quirk's test [19], steady planar shock wave problem [22] or the Double Mach Reflection (DMR) problem [8]. Due to Quirk's discovery, the stability analysis becomes very simple since it can be performed on a planar rectangular grid rather than on a complex geometry. From then on, several attempts have been made to understand and cure the phenomenon. Here we only review some viewpoints associated with the topic of this article.

One of many plausible explanations for the carbuncle phenomenon is that it is associated to an unfavorable coupling between normal and transverse directions across the shock wave. Many carbuncle cures are based on this understanding, such as Harten's entropy fix [10], Quirk's hybrid method [19] and multi-dimensional dissipation [16] [22]. In fact, most of algorithms which flag cells where the shock waves are likely to appear and make modifications are affected by the

* Corresponding author at: Institute of Applied Physics and Computational Mathematics, LCP, P. O. Box 8009-26, Beijing 100088, China.

E-mail addresses: qingdaoliuli@163.com (L. Liu), lixia0404@qq.com (X. Li), shen_zhijun@iapcm.ac.cn (Z. Shen).

viewpoint. Although the explanation provides much helps in designing numerical methods, the concept of 'unfavorable coupling' is ambiguous and provides no quantitative results.

A famous conjecture about the root of the multidimensional shock instability comes from the work of Liou [14]. He regarded that if the mass flux of a given numerical flux does not depend on the pressure term, then the scheme will not produce carbuncle solutions. Unfortunately, many counterexamples show that Liou's hypothesis is not correct [5]. In fact, those methods based on shear wave fix [24] [27] are all counterexamples of Liou's conjecture. However, Liou's work includes an inspiration that the root of the multidimensional shock instability might fully or partly come from 1D flux function. The idea is consistent with others' researches involving 1D instability [1] [12].

Dumbser et al. [5] designed a matrix stability analysis method, and stated that the carbuncle phenomenon of shock wave is not only governed by the upstream Mach number but also by the shock's internal structure. The intermediate shock points play a very key role in the onset of the instability. If the state of the intermediate point is sufficiently close to the downstream state, then carbuncle problem will not appear no matter how high the upstream Mach number. Conversely, the closer the intermediate state is to the upstream state, the more likely carbuncle behaviors would be observed. However, the finding has not yet been applied in constructing robust numerical schemes.

A very interesting point is the spatial localization of the source of instability. There are different viewpoints in this problem. Xu and Hu [28] supposed that the shock instability originates from the downstream region, while Dumbser et al. [5] regarded that it lies upstream of the shock and is convected downstream. Recently, Xie et al. [27] showed the instability comes from intermediate cell of the shock. Once more, these viewpoints are rarely taken into account in the calculations for reducing carbuncle errors.

There are many healing approaches to reduce carbuncle phenomenon. We only introduce two types of representative methods.

The first kind of cure for the carbuncle problem involves the use of 'blended' flux functions, which was first proposed by Quirk [19]. He found that approximate Riemann solvers, which exactly capture contact and shear discontinuities, generally suffer from such failings. For example, Roe's scheme and the HLLC scheme [25] are shock unstable. While high dissipative methods, such as the HLLC method [9], are insusceptible to the carbuncle problem. Using a pressure gradient to detect a shock, Quirk constructed a hybrid method by combining the high dissipative flux HLLC and low dissipative Roe scheme to suppress shock instability. In other hybrid methods such as [16] [24] [26] [27] [15], different techniques to detect cell faces where the shock wave presents and different dissipative schemes are adopted. In general, such healing has hypotheses that high dissipative schemes are always stable in calculations. If high dissipative schemes are not stable, the hybrid scheme will lose meaning. In almost all literatures, the HLLC scheme is believed to be carbuncle free.

The second class of cure is to add multidimensional dissipation. Sanders et al. [22] noticed that one-dimensional upwind schemes often lead to poor results when computing the multidimensional stationary shocks. They showed that this failure is the result of inadequate crossflow dissipation implied by strictly upwind schemes. A multidimensional dissipation based on one-dimensional entropy correction [9] is proposed to eliminate the flaws. Shen et al. [23] showed the instability is only associated with linear shear wave on the cell faces perpendicular to the shock. Inadequate shear viscosities on these faces lead to inconsistency of normal mass fluxes along the transverse direction of the shock, and trigger instability ultimately. The conclusion is consistent with the work of [27] and provides a basis to design robust numerical schemes by smearing shear wave. For example, the HLLC flux [25] and HLLC flux [7] are unstable due to fully wave resolution, but their modification versions, denoted as the HLLCM and HLLC ones respectively [24] [27], restore the stability for strong shock since shear waves are deleted completely. Other multidimensional effects are considered in rotational Riemann solver [20] [3] [13]. Essentially, the stabilities of these rotational schemes still depend on the increase of shear viscosity.

This paper investigates solution behaviors of some HLLC-type Riemann solvers for the Euler equations with the strong shock interaction. Here the HLLC-type Riemann solvers not only include original HLLC approximate flux itself [9] [6], but also include other approximate Riemann solvers stemming from it, such as the HLLC [25] [2] and its modification HLLCM scheme [24], the HLLC [7] and its modification HLLC flux [27]. A typical feature of these schemes is that their signal waves are bounded by the eigenvalues of Roe matrix and sound speed. Due to this property, these schemes can resolve isolate shock accurately and satisfy the entropy condition. However we find that all these HLLC-type numerical fluxes might suffer from carbuncle instability under certain strong shock interaction. This finding indicates that both of previous two kinds of cures could fail under some circumstances. For example, all hybrid schemes utilizing the HLLC flux as high dissipative one are exposed to the risk of instability due to behaviors of the HLLC scheme.

To better understand the onset of instability in the HLLC-type Riemann solvers, we have to revisit the source of instability and search for the key factor in influencing instability. Borrowed from the matrix stability analysis [5] [23], a stability threshold value independent of the shock strength is found to trigger instability for a given HLLC-type scheme. Once the density transition on a specific interface in the shock layer exceeds the threshold, the instability will happen. Building on the finding, we propose a simple modification method by adjusting signal velocities to heal the deficiency. For a given HLLC-type flux function on a square grid, by an analysis for viscous shock structure, a quantitative adjusting factor can be obtained and the modified scheme will be carbuncle free no matter how high the upstream Mach number is.

The outline of the paper is as follows. In Section 2, the governing equations and some Riemann solvers are introduced. An unstable phenomenon is described in Section 3, and the computing effects using two different wavespeed estimates are illustrated in the section. In Section 4, a linear matrix stability analysis is reviewed, and the stabilities of the HLLC-type schemes are discussed. Section 5 presents a carbuncle cure method for HLLC-type flux functions by adjusting signal

velocities. The emphasis of this section is to discuss the properties of viscous shock structure. Section 6 describes effect of grid aspect ratio and the solution behaviors of an inviscid flow with blunt body. The conclusions are summarized in Section 7.

2. Governing equations and HLL-type schemes

The governing equations for inviscid flow in two dimensions are as follows:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = 0, \quad (1)$$

where the state vector and flux vectors are

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho Eu + pu \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho Ev + pv \end{bmatrix},$$

where ρ, p, E are the fluid density, pressure and total energy respectively, and $\mathbf{u} = (u, v)$ is the fluid velocity. The equation of state is in the form

$$p = (\gamma - 1)\rho e = (\gamma - 1)\rho \left[E - \frac{1}{2}(u^2 + v^2) \right],$$

where γ is the specific heat ratio.

A semi-discrete finite volume method to solve (1) reads

$$\frac{d\mathbf{U}_{i,j}}{dt} = -\frac{\mathbf{H}_{i+1/2,j} - \mathbf{H}_{i-1/2,j}}{\Delta x} - \frac{\mathbf{H}_{i,j+1/2} - \mathbf{H}_{i,j-1/2}}{\Delta y}, \quad (2)$$

where the numerical fluxes $\mathbf{H}_{i+1/2,j}$ and $\mathbf{H}_{i,j+1/2}$ are approximate intercell ones of $\mathbf{F}(x_{i+1/2}, y_j)$ and $\mathbf{G}(x_i, y_{j+1/2})$ respectively. They can be obtained by many methods. In this paper, we concentrate on the HLL-type Riemann solvers.

2.1. A review of HLL-type schemes

The HLL Riemann solver is an upstream difference scheme proposed by Harten, Lax and van Leer [9], which approximates the solution of Riemann problem with two waves propagating at speeds of S_L and S_R . They should be the lower and upper bounds for the physical signal speeds with which the information of the initial discontinuity is transported. Given the initial states $\mathbf{U}_L$ and $\mathbf{U}_R$ of a Riemann problem, the numerical flux of the HLL approximate Riemann solver can be expressed as

$$\mathbf{H}_{HLL} = \frac{S_R^+ \mathbf{F}_L - S_L^- \mathbf{F}_R + S_R^+ S_L^- (\mathbf{U}_R - \mathbf{U}_L)}{S_R^+ - S_L^-}, \quad (3)$$

with $S_L^- = \min(S_L, 0)$, $S_R^+ = \max(S_R, 0)$.

To satisfy the entropy and the positivity conditions, Einfeldt et al. [6] suggested adequate bounds by making use of the Roe-averaged eigenvalues,

$$S_L = \min(u_L - a_L, \hat{u} - \hat{a}), \quad S_R = \max(u_R + a_R, \hat{u} + \hat{a}), \quad (4)$$

where a_L and a_R are the sound speeds of the left and the right states. The superscript $\hat{\cdot}$ denotes the Roe-averaged values. With the wavespeed estimates (4), the HLL flux (3) is also called HLLC one.

In most of literatures, the HLLC scheme (3) is believed to be carbuncle-free due to its high dissipation in contact discontinuity. Simultaneously, it is also regarded to be difficult to simulate practical multi-phase problems because of its this property. The HLLC-type schemes and HLLC-type schemes [7] [25] are modified versions of the HLLC scheme by restoring contact or shear waves. Therefore, we also call them as the HLLC-type schemes through out this paper.

2.2. HLLC-type schemes

The HLLC scheme [7] improves the resolution of contact discontinuity by reusing the information of contact discontinuity in terms of modifying the intermediate state. The numerical flux function of the HLLC scheme can be written as

$$\mathbf{H}_{HLLC} = \mathbf{H}_{HLL} - \frac{S_R^+ S_L^-}{S_R^+ - S_L^-} (\delta_2 \hat{\alpha}_2 \hat{\mathbf{r}}_2 + \delta_3 \hat{\alpha}_3 \hat{\mathbf{r}}_3), \quad (5)$$

where $\hat{\mathbf{r}}_k$ correspond the right eigenvectors of the flux Jacobian $\hat{\mathbf{A}}$ evaluated at the intermediate states $\hat{\mathbf{U}}$. The wave strengths $\hat{\alpha}_k$ are the approximate values of the projection from $\mathbf{U}_R - \mathbf{U}_L$ onto $\hat{\mathbf{r}}_k$, i.e.,

$$\mathbf{U}_R - \mathbf{U}_L = \sum_{k=1}^4 \hat{\alpha}_k \hat{\mathbf{r}}_k. \quad (6)$$

δ_2 and δ_3 are the anti-diffusion coefficients to take out excess dissipation in the linear degenerate fields [6],

$$\delta_2 = \delta_3 = \frac{\hat{a}}{|\bar{u}| + \hat{a}}, \quad (7)$$

where the speed $\bar{u}$ is defined as the approximate speed of the contact discontinuity, which is originally obtained by satisfying stability condition of HLLEM scheme [6]. Here we set $\bar{u} = \hat{u}$ to enhance the resolution of linear degenerate waves [17].

Letting the coefficient of shear wave $\delta_3 = 0$ and keeping the contact wave unchanged, the HLLEM flux function (5) degenerates to the HLLEC one [27],

$$\mathbf{H}_{HLLEC} = \mathbf{H}_{HLLE} - \frac{S_R^+ S_L^-}{S_R^+ - S_L^-} \delta_2 \hat{\alpha}_2 \hat{\mathbf{r}}_2. \quad (8)$$

The scheme HLLEC is regarded as free from shock instability, refer to [27].

2.3. HLLC-type schemes

The HLLC Riemann solver [25] is considered to be the most preferred method because of its simplicity and ability to capture discontinuities accurately. In the solver, four constant states, which can be denoted from left to right as $\mathbf{U}_L, \mathbf{U}_L^*, \mathbf{U}_R^*, \mathbf{U}_R$, are separated by left signal velocity S_L , contact velocity S_* and right signal velocity S_R respectively. The flux function can be expressed as

$$\mathbf{H}_{HLLC} = \begin{cases} \mathbf{F}_L, & \text{if } S_L \geq 0, \\ \mathbf{F}_L + S_L(\mathbf{U}_{HLLC,L}^* - \mathbf{U}_L), & \text{if } S_L < 0 \leq S_*, \\ \mathbf{F}_R + S_R(\mathbf{U}_{HLLC,R}^* - \mathbf{U}_R), & \text{if } S_* < 0 < S_R, \\ \mathbf{F}_R, & \text{if } S_R \leq 0, \end{cases} \quad (9)$$

where the conserved vectors in the star region are

$$\mathbf{U}_{HLLC,K}^* = \rho_K^* \begin{bmatrix} 1 \\ S_* \\ v_K \\ E_K^* \end{bmatrix} = \rho_K^* \begin{bmatrix} 1 \\ S_* \\ v_K \\ E_K^{*,1d} + \frac{1}{2} v_K^2 \end{bmatrix}, \quad \text{for } K = L, R, \quad (10)$$

in which the total energy is decomposed into sums of a pure one-dimensional normal quantity and a tangent kinetic energy. The contact velocity in the star region is

$$S_* = \frac{\rho_R(S_R - u_R)u_R - \rho_L(S_L - u_L)u_L + p_L - p_R}{\rho_R(S_R - u_R) - \rho_L(S_L - u_L)}. \quad (11)$$

Introduce the following notations [24],

$$\alpha_L = \rho_L(S_L - u_L), \quad \alpha_R = \rho_R(S_R - u_R),$$

then the density and total energy in the star region are

$$\rho_K^* = \frac{\alpha_K}{S_K - S_*}, \quad E_K^{*,1d} = e_K + \frac{1}{2} u_K^2 + (S_* - u_K) \left(S_* + \frac{p_K}{\alpha_K} \right), \quad K = L, R. \quad (12)$$

In order to improve stability, one introduces the shear viscosity [24] to the HLLC scheme. The new algorithm is called HLLCM solver and has same flux form with (9),

$$\mathbf{H}_{HLLCM} = \begin{cases} \mathbf{F}_L, & \text{if } S_L \geq 0, \\ \mathbf{F}_L + S_L(\mathbf{U}_{HLLCM,L}^* - \mathbf{U}_L), & \text{if } S_L < 0 \leq S_*, \\ \mathbf{F}_R + S_R(\mathbf{U}_{HLLCM,R}^* - \mathbf{U}_R), & \text{if } S_* < 0 < S_R, \\ \mathbf{F}_R, & \text{if } S_R \leq 0. \end{cases} \quad (13)$$

Here, the conserved vectors in star states are

$$\mathbf{U}_{HLLCM,K}^* = \rho_K^* \begin{bmatrix} 1 \\ S_* \\ v^* \\ E_{HLLCM,K}^* \end{bmatrix} = \rho_K^* \begin{bmatrix} 1 \\ S_* \\ \frac{\alpha_R v_R - \alpha_L v_L}{\alpha_R - \alpha_L} \\ E_K^{*,1d} + \frac{1}{2} \frac{\alpha_R v_R^2 - \alpha_L v_L^2}{\alpha_R - \alpha_L} \end{bmatrix}, \quad \text{for } K = L, R. \quad (14)$$

The main difference between the HLLC and HLLCM is in the tangent velocity. The change of total energy is along with the variation of tangent velocity. The latter Riemann solver is also a very robust scheme [24].

3. Some instability phenomena

In what follows, we use numerical experiments to illustrate some solution behaviors of these HLLC-type schemes.

The raw problem is a steady normal 1-shock wave. The initial data are given by the exact Rankine-Hugoniot solution in x -direction. The upstream states are

$$\mathbf{W}_L = (\rho_L, u_L, v_L, p_L) = (1, \sqrt{\gamma} M_a, 0, 1), \quad x < L/2, \quad (15)$$

and the downstream states are

$$\mathbf{W}_R = (\rho_R, u_R, v_R, p_R) = \left(\rho_L f(M_a), \frac{u_L}{f(M_a)}, 0, p_L g(M_a) \right), \quad x > L/2, \quad (16)$$

where $M_a = u_L/a_L$ is the upstream Mach number of the flow field, and a_L is the sound speed,

$$f(M_a) = \left(\frac{2}{\gamma+1} \frac{1}{M_a^2} + \frac{\gamma-1}{\gamma+1} \right)^{-1}, \quad g(M_a) = \frac{2\gamma}{\gamma+1} M_a^2 - \frac{\gamma-1}{\gamma+1}.$$

In our calculations, the specific heat ratio $\gamma = 1.4$, Mach number $M_a = 20$. The computational domain is $[0, L] \times [0, 1]$, here $L = 0.2$ and the mesh has 20×100 uniform cells. Einfeldt et al.'s algorithm (4) is adopted to estimate signal velocities. A slight random perturbation of the relative order 10^{-15} is added to the downstream states. The upper and bottom boundary conditions are the outflow. The boundary conditions of the left and right side are the inflow and outflow respectively.

The density contours using the schemes of HLEM, HLLC, HLLC, HLLCM and HLLC are displayed in Fig. 1. The shock positions in all schemes move and thus deviate from the correct solution. Some move to the left and some to the right. Unlike other flux functions, the HLLC scheme does not display sawtooth forms of the unstable mode at the shock surface. However, further observation, by plotting physical quantities in different time, shows that the typical spatial sawtooth patterns always appear at certain stages for a scheme with wrong shock location. Thus we still call the shifting phenomena in the HLLC scheme as carbuncle ones. These results show the above HLLC-type schemes suffer from carbuncle instability.

It is well known that the HLEM flux and HLLC flux are both susceptible to the carbuncle problem since they are both able to exactly capture the contact and shear discontinuities. But the solution behaviors in the HLLC, HLLCM and HLLC fluxes are a little surprising. The shear viscosities in these schemes have been manipulated to the largest extent and provide a kind of multi-dimensional dissipative mechanism to depress instability. The three schemes should have more stable behaviors while they actually don't have.

We have to try some ways to increase viscosity in normal direction of shock wave and observe the influence to stability of numerical solution. In this paper we change viscosity by adjusting signal velocities of the HLLC-type schemes.

3.1. Davis's wavespeed estimates

Davis's wavespeed algorithm [4] is the simplest one

$$S_L = \min(u_L - a_L, u_R - a_R), \quad S_R = \max(u_L + a_L, u_R + a_R). \quad (17)$$

In general, it is regarded more diffusive than Einfeldt et al.'s wave speed estimates (4) since it can not distinguish an isolated shock. The numerical results using Davis's wave speed estimates (17) are different from those in Fig. 1. As shown in Fig. 2, the HLEM and HLLC fluxes still suffer from catastrophic instability, while anomalous phenomena in the fluxes of the HLLC, HLLCM and HLLC disappear. The latter three results have correct shock position. In order to distinguish a flux with Davis's algorithm from Einfeldt et al.'s algorithm, we add a subscript 'D' to the flux.

One question arises naturally here: is Davis's algorithm a true cure to carbuncle deficiencies for HLLC-type schemes? If so, what is the cure mechanism of Davis's algorithm? To answer this question, we need to perform a stability analysis.

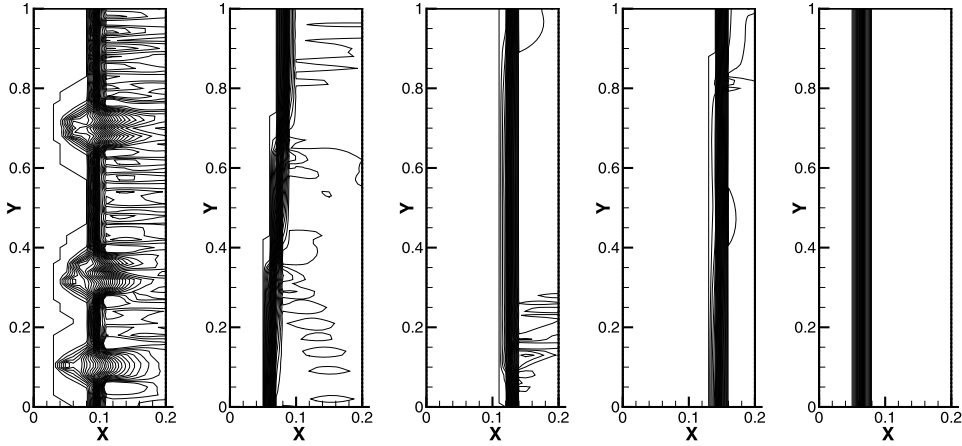


Fig. 1. The density contours for 2D steady flow with Einfeldt's wave velocity estimate. The grid resolution is 20×100 cells. Thirty-one equally spaced contour lines from $\rho = 1$ to $\rho = 6$. From left to right: *HLLEM*, *HLLC*, *HLLC*, *HLLCM*, *HLLC*. All of them are not stable.

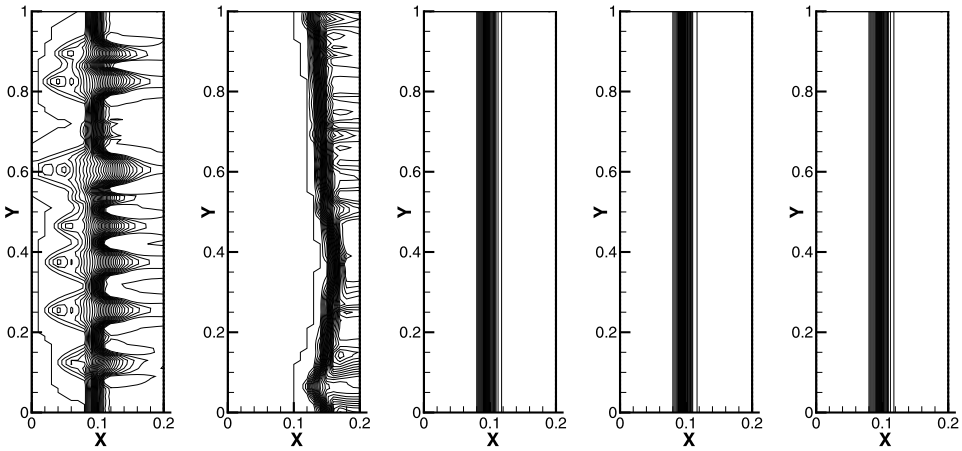


Fig. 2. The density contours for 2D steady flow with Davis' wave velocity estimate. The grid resolution is 20×100 cells. Thirty-one equally spaced contour lines from $\rho = 1$ to $\rho = 6$. From left to right: *HLLC_D*, *HLLC_D*, *HLLC_D*, *HLLC_D*, *HLLC_D*.

4. Stability analysis

In a 1D stationary shock calculation, captured shock frequently contains some intermediate states, as shown in Fig. 3. They are not part of the exact solution but products of the discrete representation when numerical scheme driving the system toward equilibrium. And the number of required intermediate states depends on different schemes and initial data. For example, Einfeldt et al.'s algorithm usually keeps $0 \sim 1$ transition cell, and Davis's algorithm includes more transition cells. For a given numerical scheme, the discrete steady distributions on these intermediate cells construct a family of stationary solutions.

In this paper, we always assume that the upstream and (limit) downstream states of a stationary shock are (15) and (16) respectively. Their conservative quantities are denoted as $\mathbf{U}_L$ and $\mathbf{U}_R$ respectively.

4.1. A short review of matrix stability analysis

A matrix-based stability analysis [5] [23] has been used to study the occurrence of unstable modes during the shock wave computation. Here we describe briefly the approach in [5] [23] for the convenience of discussion.

Similar to numerical experiments in preceding section, calculations are performed on a 2D domain $[0, 1] \times [0, 1]$. The grid is composed of regular Cartesian cells without perturbation. The total cell number is $M = I \times J$.

For the stability analysis of a steady field, we assume

$$\mathbf{U}_{i,j} = \mathbf{U}_{i,j}^0 + \delta \mathbf{U}_{i,j}, \quad (18)$$

where $\mathbf{U}_{i,j}^0$ is the steady (initial) conserved vector and $\delta \mathbf{U}_{i,j}$ is small perturbation.

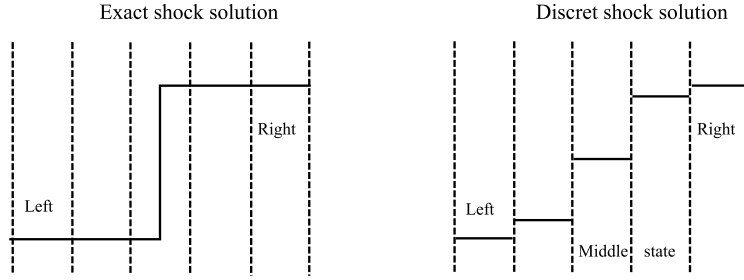


Fig. 3. A representative plot of a single shock problem. Left: initial state, Right: steady state.

Substituting (18) into (2), and the numerical flux functions are linearized around the steady mean value, then we finally get the error evolution of all M cells in the domain,

$$\frac{d}{dt} \begin{pmatrix} \delta \mathbf{U}_{1,1} \\ \vdots \\ \delta \mathbf{U}_{I,J} \end{pmatrix} = S \begin{pmatrix} \delta \mathbf{U}_{1,1} \\ \vdots \\ \delta \mathbf{U}_{I,J} \end{pmatrix}, \quad (19)$$

where S is the stability matrix. Refer to [5] and [23] for details. When considering only the evolution of initial errors, the solution of the linear system (19) is

$$\begin{pmatrix} \delta \mathbf{U}_{1,1} \\ \vdots \\ \delta \mathbf{U}_{I,J} \end{pmatrix} = \exp^{St} \begin{pmatrix} \delta \mathbf{U}_{1,1}^0 \\ \vdots \\ \delta \mathbf{U}_{I,J}^0 \end{pmatrix}. \quad (20)$$

The perturbation remains bounded only if the maximum of the real part of the eigenvalues of S is negative,

$$\max(\operatorname{Re}(\lambda)) \leq 0. \quad (21)$$

For a specific numerical scheme, this stability analysis needs a pure one-dimensional numerical stationary shock as background solution $\mathbf{U}_{i,j}^0$.

4.2. Single point equilibrium state in Einfeldt et al.'s algorithm

In literature [1], Barth proved that there exists a one-parameter family stationary shocks for Godunov method and Roe method, in which the intermediate state denoted by $\mathbf{U}_m$, is set on the Hugoniot curve based on the downstream state $\mathbf{U}_R$. Here we write the intermediate states as

$$\begin{aligned} \rho_m &= \alpha_\rho \rho_L + (1 - \alpha_\rho) \rho_R, \quad p_m = \frac{(\gamma + 1)\rho_m - (\gamma - 1)\rho_R}{(\gamma + 1)\rho_R - (\gamma - 1)\rho_m} p_R, \\ u_m &= u_R + (p_R - p_m) \left[\frac{2/\rho_R}{(\gamma - 1)p_R + (\gamma + 1)p_m} \right]^{1/2}, \quad v_m = 0, \end{aligned}$$

where the parameter $0 < \alpha_\rho \leq 1$. The state is on the I_m -th cell, and initial values are

$$\mathbf{U}_{i,j}^0 = \begin{cases} \mathbf{U}_L, & \text{if } i < I_m, \quad j = 1, \dots, J, \\ \mathbf{U}_m, & \text{if } i = I_m, \quad j = 1, \dots, J, \\ \mathbf{U}_R, & \text{if } i > I_m, \quad j = 1, \dots, J. \end{cases} \quad (22)$$

Note that in pure 1D circumstance, there is always

$$HLLEC = HLLEM \quad \text{and} \quad HLLCM = HLLC. \quad (23)$$

Theorem 1. The initial distribution of (22) is a steady shock state of HLE-type Riemann solvers. The interfaces of $(\mathbf{U}_L, \mathbf{U}_m)$ and $(\mathbf{U}_m, \mathbf{U}_R)$ satisfy respectively

$$\begin{cases} S_L(\mathbf{U}_L, \mathbf{U}_m) > 0, \\ S_R(\mathbf{U}_L, \mathbf{U}_m) > 0, \end{cases} \quad \text{and} \quad \begin{cases} S_L(\mathbf{U}_m, \mathbf{U}_R) < 0, \\ S_R(\mathbf{U}_m, \mathbf{U}_R) > 0. \end{cases} \quad (24)$$

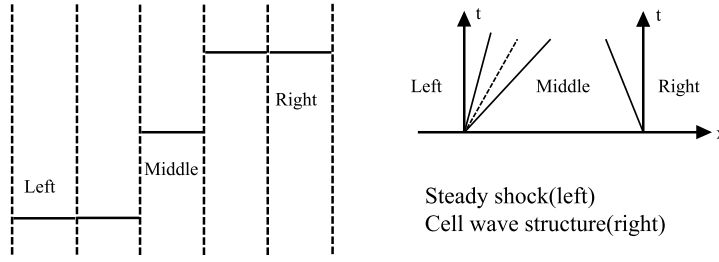


Fig. 4. A single equilibrium steady shock and the cell wave structures for Einfeldt et al.'s algorithm.

Proof. Obviously, the intermediate state satisfies

$$\rho_L < \rho_m < \rho_R, \quad p_L < p_m < p_R, \quad u_L > u_m > u_R > 0, \quad v_L = v_m = v_R = 0. \quad (25)$$

Firstly let's consider the interface of $(\mathbf{U}_m, \mathbf{U}_R)$. According to definition of $\mathbf{U}_m$, the Riemann solution between $\mathbf{U}_m$ and $\mathbf{U}_R$ is a left-moving single shock with speed $s < 0$, and shock relations satisfy

$$\mathbf{F}_R - \mathbf{F}_m = s(\mathbf{U}_R - \mathbf{U}_m), \quad u_m - a_m > s > u_R - a_R. \quad (26)$$

For such an isolated 1-shock, Roe decomposition (6) satisfies [21]

$$\hat{u}(\mathbf{U}_m, \mathbf{U}_R) - \hat{a}(\mathbf{U}_m, \mathbf{U}_R) = s < 0, \quad \hat{\alpha}_2(\mathbf{U}_m, \mathbf{U}_R) = \hat{\alpha}_3(\mathbf{U}_m, \mathbf{U}_R) = \hat{\alpha}_4(\mathbf{U}_m, \mathbf{U}_R) = 0.$$

This implies that all wave strengths in Roe decomposition are zero except the 1-wave. According to the definition of Einfeldt et al.'s algorithm and (26), we have

$$S_L(\mathbf{U}_m, \mathbf{U}_R) = s < 0, \quad S_R(\mathbf{U}_m, \mathbf{U}_R) \geq u_R + a_R > 0.$$

Substituting these relations into (3), (5), (8), one can obtain

$$\mathbf{H}_{HLLM} = \mathbf{H}_{HLLC} = \mathbf{H}_{HLL} = \frac{S_R \mathbf{F}_m - s \mathbf{F}_R + S_R s (\mathbf{U}_R - \mathbf{U}_m)}{S_R - s} = \mathbf{F}_R.$$

For the HLLC scheme, from (11), we have

$$S_*(\mathbf{U}_m, \mathbf{U}_R) = \frac{\rho_R(S_R - u_R)u_R - \rho_m(s - u_m)u_m + p_m - p_R}{\rho_R(S_R - u_R) + \rho_m(s - u_m)} = \frac{\rho_R u_R (S_R - s)}{\rho_R (S_R - s)} = u_R.$$

Thus there are

$$\mathbf{U}_K^*(\mathbf{U}_m, \mathbf{U}_R) = \mathbf{U}_R, \quad \text{for } K = L, R.$$

Furthermore,

$$\mathbf{H}_{HLLCM} = \mathbf{H}_{HLLC} = \mathbf{F}_R.$$

Secondly we consider the interface of $(\mathbf{U}_L, \mathbf{U}_m)$. Barth [1] proved the interface of $(\mathbf{U}_L, \mathbf{U}_m)$ is supersonic for the exact Godunov flux and Roe flux. Especially, 1-wave is a shock wave. This means:

$$\hat{u}(\mathbf{U}_L, \mathbf{U}_m) - \hat{a}(\mathbf{U}_L, \mathbf{U}_m) > 0. \quad (27)$$

Once more, according to the definition of Einfeldt et al.'s algorithm, there is

$$S_L(\mathbf{U}_L, \mathbf{U}_m) > 0.$$

The proof is completed. The wave structures on the two interfaces are illustrated in Fig. 4. The one-parameter family initial value $(\mathbf{U}_L, \mathbf{U}_m, \mathbf{U}_R)$ keeps unchanged when using the HLLC-type schemes to update.

In this work, a cell boundary is termed transonic interface if it is between a supersonic cell front ($S_L > 0$ or $S_R < 0$) and a subsonic cell front ($S_L \leq 0 \leq S_R$). Here the interface of $(\mathbf{U}_m, \mathbf{U}_R)$ is transonic.

Table 1

The stable threshold values α_{thr} for Einfeldt et al.'s algorithm in 1D case.

Mach	HLLEM	HLLC	HLLEC	HLLCM	HLLE
10	0.980	0.980	0.980	0.980	0.980
20	0.973	0.973	0.973	0.973	0.973
30	0.971	0.971	0.971	0.971	0.971
...					
100	0.970	0.970	0.970	0.970	0.970

Table 2

The stable threshold values α_{thr} for Einfeldt et al.'s algorithm in 2D case.

Mach	HLLEM	HLLC	HLLEC	HLLCM	HLLE
10	0.380	0.380	0.816	0.816	0.935
20	0.375	0.375	0.810	0.810	0.928
30	0.374	0.374	0.809	0.809	0.927
...					
100	0.373	0.373	0.808	0.808	0.926

4.3. Analysis results to the HLLE-type Riemann solvers

Given a numerical steady shock, one may carry out stability analysis for all HLLE-type Riemann solvers. The analysis method is applicable for both 1D and 2D problems. In fact, 1D analysis is a natural degeneration of 2D analysis, in which we keep fluxes in y-direction constant.

When varying the intermediate state by adjusting α_ρ , the stabilities of these solutions to perturbations are different. Just as pointed out by Dumbser et al. in [5], the stability is

$$\begin{cases} \text{as } \alpha_\rho \rightarrow 1 (\rho_m \rightarrow \rho_L), & \text{unstable,} \\ \text{as } \alpha_\rho \rightarrow 0 (\rho_m \rightarrow \rho_R), & \text{stable.} \end{cases}$$

Therefore, there exists a threshold value of α_ρ , which is the upper bound of stable region (or lower bound of unstable region). Denote the threshold as α_{thr} , then stable domain of a flux function can be expressed as $(0, \alpha_{thr}]$, and unstable domain is written as $(\alpha_{thr}, 1]$.

Analysis results indicate all flux formulae have unstable regions for strong shock problems. With the increase of shock strength (represented by upstream Mach number), α_{thr} decreases. This means the scheme is more susceptible to destabilizing. The stable threshold values α_{thr} for strong shocks are listed in the Table 1 and Table 2, corresponding to 1D and 2D cases respectively.

The following facts can be observed from Table 1 and Table 2:

1. For pure 1D cases, unstable regions are very small. This is easy to understand since perturbations in y-direction are neglected. Viewed from these results, the linear contact waves on the interface have very little effect, therefore all HLLE-type fluxes have the same stability region with two-wave HLLE flux.

2. For 2D cases, unstable regions become larger than 1D cases due to more perturbation factors. Shear viscosity plays a very important role in enhancing stability. The HLLCM and HLLEC fluxes have better stability than the HLLC and HLLEM fluxes, whereas the HLLE scheme has best stability. Note that the HLLE scheme has the same shear viscosity as the HLLCM flux but different contact wave [24], this fact shows that the linear contact wave has important influence to trigger shock instability too.

From the definition of intermediate density ρ_m and initial values (15) and (16), we know

$$\frac{\rho_m}{\rho_R} = 1 - \alpha_\rho + \alpha_\rho \frac{\rho_L}{\rho_R} = 1 - \frac{2\alpha_\rho}{\gamma + 1} \left(1 - \frac{1}{M_a^2} \right). \quad (28)$$

Substituting threshold value α_{thr} in Table 1 and 2 into above formula, an interesting finding is that:

Proposition 1. For a given HLLE-type scheme, there exists a roughly constant value C_{thr} independent of shock intensity so that when density ratio ρ_R/ρ_m is larger than the constant number, instability will occur. In other word, if $\rho_R/\rho_m < C_{thr}$, the calculation is carbuncle free no matter how high the upstream Mach number is.

Here $C_{thr} = \rho_R/\rho_m(\alpha_{thr})$. Table 3 lists these threshold values for different flux functions, in which subscript of $C_{thr,i}$ corresponds to different scheme respectively.

Proposition 1 can be understood as a quantitative extension to work of Dumbser et al. [5]. There are following remarks for Proposition 1:

Table 3Density jump threshold ρ_R/ρ_m of instability for Einfeldt et al.'s algorithm.

Mach	1D C_{1d}	HLLEM and HLLC $C_{thr,1}$	HLLEC and HLLCM $C_{thr,2}$	HLLE $C_{thr,3}$
10	5.22	1.457	3.060	4.374
20	5.23	1.452	3.061	4.374
30	5.22	1.452	3.062	4.379
...				
100	5.22	1.451	3.061	4.378
100		Numerically	2.78	4.1

Remark 1. The linear matrix analysis is very good theoretical prediction to practical calculations. It is because the eigenvalues of Jacobian matrix of the HLLE-type flux functions are non-zero in this single point equilibrium distribution and so these fluxes become differentiable around the mean flow. However, there still exist some deviations between the linear analysis and nonlinear behavior obtained in a numerical experiment. In practical calculations, $C_{thr,2} \approx 2.78$, $C_{thr,3} \approx 4.1$. Refer to Table 3.

Remark 2. The analysis consequents in Table 1 and 2 are valid for general initial values. For example, if we change velocity and pressure in (15) as follows [5],

$$\mathbf{W}_L = (\rho_L, u_L, v_L, p_L) = (1, 1, 0, \gamma M_a^2), \quad (29)$$

then analysis results are the same as those in Table 1 and 2. However, in practical calculations, the onset time of instability varies in a large range. The flowfield destabilizes at about $t = 2$ for initial value (15), but at $t = 40$ or so for (29). It is because the carbuncle phenomena appear with the accumulation of numerical errors which depend on the number of time steps rather than physical time. Time steplength of (15) is much less than that of (29).

Remark 3. In general, the stability depends on numerical shock structure unless shock strength is so small that density ratio $\rho_R/\rho_L < C_{thr}$. The latter is called unconditional stability which is irrelevant to shock structure. For a practical strong shock problem, one does not know numerical shock structures in advance, therefore an unstable risk always exists during the calculation.

The conclusions in Proposition 1 are obtained from a planar stationary shock and a square grid. However it still can provide help to understand more carbuncle phenomena.

4.4. A description of the evolution procedure of carbuncle instability

In the above matrix analysis process, one can not distinguish shock interface and other interfaces. In terms of Proposition 1, magnitude of density jump is a necessary condition to cause instability. It is not a sufficient condition because there are two possible large density ratios ρ_R/ρ_m and ρ_m/ρ_L , but stability only depends on the former. This fact leads us to believe that the interface of $(\mathbf{U}_m, \mathbf{U}_R)$ is spatial localization of the source of instability. Notice that the main difference between the interfaces of $(\mathbf{U}_m, \mathbf{U}_R)$ and $(\mathbf{U}_L, \mathbf{U}_m)$ is in their wave structures. The cell edge $(m - 1/2, j)$ is a supersonic front with all right-traveling waves, while the interface $(m + 1/2, j)$ includes a left-traveling isolated 1-wave (shock), and three right-traveling waves with zero amplitude, refer to Fig. 4. Once small disturbances are introduced into these interfaces, disturbance waves will send wave information to neighboring cells to drive the system toward new equilibrium. During the procedure, only the left-traveling wave with large jump on $(m + 1/2, j)$ can amplify perturbations passing through it.

For a 2D stationary shock, amplified perturbations can result in nonuniform states on cells (m, j) and $(m + 1, j)$ in the transverse direction of shock wave. Such inconsistency is the root of spatial sawtooth patterns along the shock surface. Table 2 tells us the linear waves (2-wave and 3-wave) play important roles in amplifying discrepancies along y-direction. Hence the contact-resolving HLLE-type fluxes are less stable than the HLLE one,

$$C_{thr,3} > C_{thr,1}, \quad C_{thr,3} > C_{thr,2}.$$

The HLLCM and HLLEC fluxes smear shear wave (3-wave) and thus have better stability than those shear-resolving schemes, i.e., $C_{thr,2} > C_{thr,1}$. In fact, the effect of shear viscosity is mainly confined to cell face perpendicular to the shock [23], therefore it is conducive to keep coherent normal momentum ρu along the transverse direction of shock wave and naturally damp out some instability.

Proposition 1 shows that the role of shear viscosity is limited in keeping consistent distribution in y-direction. If intrinsic density jump ρ_R/ρ_m is greater than $C_{thr,2}$ or $C_{thr,3}$, the shear viscosity is inadequate to suppress perturbations and a sawtooth form of the shock profile might appear.

Remark 4. If a small disturbance appears in a cell far from the shock interface, for example, a spurious reflection at an outflow boundary, it does not influence stability directly till the disturbance is propagated to the shock interface by sound waves. Refining grid or increasing cell number in the normal direction of the shock wave may damp out disturbances and thus enhance stability. On the contrary, refining grid or increasing cell number in the shear direction of the shock wave will be more likely to introduce instability.

5. Cures for the HLLC-type schemes

According to Remark 3, we know caruncle instability is hard to avoid when simulating shock problems by use of the HLLC-type schemes with Einfeldt's algorithm. It is because the density ratio on a specific cell interface is irrelevant to the numerical schemes. In this section, a simple cure method is proposed by use of justifying signal velocities of Riemann solvers and properties of modified schemes are discussed. Firstly, we focus on behaviors of an one-dimensional HLLC-type flux with Davis's algorithm and prove that it produces a viscosity shock and decrease the density difference between two adjacent cells. Secondly, we seek the interface which most influences stability and discuss its stability. Finally, we extend the research results in Davis's algorithm to more general wave velocity algorithm.

5.1. Multi-points equilibrium state in Davis's algorithm

In contrast with Einfeldt et al.'s wave algorithm, Davis's wave speed estimates can not preserve a stationary shock with single-point equilibrium state.

Theorem 2. Between a physical stationary shock $(\mathbf{U}_L, \mathbf{U}_R)$, there exists no intermediate state $\mathbf{U}_m \neq \mathbf{U}_R$ satisfying $\mathbf{H}(\mathbf{U}_m, \mathbf{U}_R) = \mathbf{F}_R$ for all HLLC_D-type flux functions.

Proof. Assume that such state $\mathbf{U}_m$ exists and satisfies (25), we prove that the wave structure at the interface of $(\mathbf{U}_m, \mathbf{U}_R)$ must be an isolated left-moving wave.

Since $\mathbf{U}_R$ is subsonic ($u_R - a_R < 0$, $u_R + a_R > 0$), it follows that the interface of $(\mathbf{U}_m, \mathbf{U}_R)$ is also subsonic, i.e., $S_L(\mathbf{U}_m, \mathbf{U}_R) < 0$ and $S_R(\mathbf{U}_m, \mathbf{U}_R) > 0$. Notice the flux $\mathbf{H}(\mathbf{U}_m, \mathbf{U}_R) = \mathbf{F}_R$, the amplitudes of the right-moving waves are identically zero and thus the jump from $\mathbf{F}_m$ to $\mathbf{F}_R$ comes from the left-moving waves.

Obviously the flux HLLC_D only includes single left-moving wave. For the HLLC_D (or HLLC_D) flux, according to (5), there is

$$\delta_2 \hat{\alpha}_2 \hat{\mathbf{r}}_2 + \delta_3 \hat{\alpha}_3 \hat{\mathbf{r}}_3 = 0.$$

This means that the intermediate waves have zero amplitude and the HLLC_D flux degenerates to the HLLC_D. As to the HLLC_D (or HLLC_D) scheme, since $\mathbf{U}_R^* = \mathbf{U}_R$, the star region has propagation speed $S_* = u_R > 0$. This implies that linear waves are right-moving. In summary, the left-moving wave is a single wave and satisfies

$$\mathbf{F}_R - \mathbf{F}_m = S_L(\mathbf{U}_R - \mathbf{U}_m).$$

This shows that the single left-moving wave is a 1-shock with velocity $s = S_L$. On the one hand, the shock speed should satisfy

$$u_m - a_m > S_L > u_R - a_R.$$

On the other hand, according to definition of Davis's algorithm, there is

$$S_L = \min(u_m - a_m, u_R - a_R) \leq u_R - a_R.$$

The two results are contradicted. This contradiction arises from single equilibrium assumption.

According to this theorem, the number of grid cells in post-shock domain should be infinity to attain limit state $\mathbf{U}_R$. Denote grid cells behind the shock as $m1, m2, \dots$, and corresponding conservative quantities are $\mathbf{U}_{m1}, \mathbf{U}_{m2}, \dots$, in which $m1$ is the first cell adjacent to $\mathbf{U}_L$ but its state is different from $\mathbf{U}_L$ ($\mathbf{U}_{m1} \neq \mathbf{U}_L$). The stationary fluxes satisfy

$$\mathbf{F}_L = \mathbf{H}(\mathbf{U}_L, \mathbf{U}_{m1}) = \mathbf{H}(\mathbf{U}_{m1}, \mathbf{U}_{m2}) = \dots = \mathbf{F}_R, \quad (30)$$

and the primitive variables satisfy

$$\begin{aligned} \rho_L &\leq \rho_{m1} \leq \rho_{m2} \leq \dots \leq \rho_R, \\ p_L &\leq p_{m1} \leq p_{m2} \leq \dots \leq p_R, \\ u_L &\geq u_{m1} \geq u_{m2} \geq \dots \geq u_R. \end{aligned} \quad (31)$$

Remark 5. The discrete stationary distribution can be obtained by a pure one-dimensional numerical calculation, provided downstream grid number is large enough. Although the states behind the shock can not equal exactly $\mathbf{U}_R$, the large deviations from $\mathbf{U}_R$ only concentrate upon a few grid cells. These few grids constitute a transition zone, or shock layer. Numerical experiments show that contact resolving $HLLD$ -type schemes include 3 ~ 5 transition cells (see Fig. 3), while downstream states outside transition zone have very slight errors with $\mathbf{U}_R (< 10^{-8})$.

Connecting stationary shock states $\mathbf{U}_L$ and $\mathbf{U}_R$, there are different discrete distributions due to variation of $\mathbf{U}_{m1}$. These numerical solutions can be obtained by adjusting the initial intermediate state $\mathbf{U}_m = \alpha_\rho \mathbf{U}_L + (1 - \alpha_\rho) \mathbf{U}_R$, that is, adjusting parameter α_ρ from 0 to 1. In general, $\mathbf{U}_{m1}$ is not equal to initial value $\mathbf{U}_m$ due to the evolution of the steady state, even the interface of $(\mathbf{U}_L, \mathbf{U}_{m1})$ might deviate from the interface $(m - 1/2, j)$.

Theorem 3. With Davis's algorithm, the interface of $(\mathbf{U}_{m1}, \mathbf{U}_{m2})$ is a transonic front, namely,

$$\begin{cases} S_L(\mathbf{U}_L, \mathbf{U}_{m1}) \geq 0, \\ S_R(\mathbf{U}_L, \mathbf{U}_{m1}) > 0, \end{cases} \quad \begin{cases} S_L(\mathbf{U}_{m1}, \mathbf{U}_{m2}) \leq 0, \\ S_R(\mathbf{U}_{m1}, \mathbf{U}_{m2}) > 0, \end{cases} \quad \text{and} \quad \begin{cases} S_L(\mathbf{U}_{m2}, \mathbf{U}_{m3}) \leq 0, \\ S_R(\mathbf{U}_{m2}, \mathbf{U}_{m3}) > 0. \end{cases} \quad (32)$$

Proof. At first, we consider the interface of $(\mathbf{U}_L, \mathbf{U}_{m1})$. Assume the state $\mathbf{U}_{m1}$ is subsonic ($u_{m1} - a_{m1} < 0$), then the interface is subsonic because $S_L(\mathbf{U}_L, \mathbf{U}_{m1}) = u_{m1} - a_{m1} < 0$. Notice that the numerical flux $\mathbf{H}(\mathbf{U}_L, \mathbf{U}_{m1}) = \mathbf{F}_L$, then the left-traveling wave in the Riemann solver has zero amplitude and right-moving waves jump from $\mathbf{F}_L$ to $\mathbf{F}_{m1}$. An argument similar to the one in Theorem 2 shows that right-moving waves include two linear waves and a nonlinear wave. Without loss of generality, we only discuss the case of the $HLLC_D$ flux. Its wave decomposition is

$$\mathbf{F}_L - \mathbf{F}_{m1} = S_*(\mathbf{U}_L - \mathbf{U}_{m1}^*) + S_R(\mathbf{U}_{m1}^* - \mathbf{U}_{m1}),$$

where $\mathbf{U}_{m1}^*$ is right star state, in which $p_{m1}^* = p_L$, $u_{m1}^* = S_* = u_L$.

After some rearrangement we obtain,

$$p_L - p_{m1} = \rho_{m1} [S_R(\mathbf{U}_L, \mathbf{U}_{m1}) - u_{m1}] (u_L - u_{m1}).$$

From $S_R(\mathbf{U}_L, \mathbf{U}_{m1}) \geq u_{m1} + c_{m1}$ and $u_L \geq u_{m1}$, one deduces that $p_L \geq p_{m1}$. This leads to a contradiction to the requirement (31). The contradictory arises from the assumption to subsonic state of $\mathbf{U}_{m1}$. We arrive at the conclusion that both the state of $\mathbf{U}_{m1}$ and interface of $(\mathbf{U}_L, \mathbf{U}_{m1})$ are supersonic and there is $u_{m1} - a_{m1} \geq 0$.

Secondly, we discuss the interface of $(m1, m2)$. Suppose the state $\mathbf{U}_{m2}$ is supersonic ($u_{m2} - a_{m2} > 0$), then the interface is also right traveling supersonic, i.e.,

$$S_L(\mathbf{U}_{m1}, \mathbf{U}_{m2}) = \min(u_{m1} - a_{m1}, u_{m2} - a_{m2}) \geq 0.$$

Then the stationary numerical flux on the cell face should be

$$\mathbf{H}(\mathbf{U}_{m1}, \mathbf{U}_{m2}) = \mathbf{F}(\mathbf{U}_{m1}) = \mathbf{F}_R = \mathbf{F}_R(\mathbf{U}_R).$$

It shows that $\mathbf{U}_{m1}$ constructs a stationary prior-shock state of $\mathbf{U}_R$ and thus $\mathbf{U}_{m1} = \mathbf{U}_L$. It contradicts the assumption $\mathbf{U}_{m1} \neq \mathbf{U}_L$. Consequently, the state $\mathbf{U}_{m2}$ is subsonic ($u_{m2} - a_{m2} \leq 0$). Furthermore,

$$S_L(\mathbf{U}_{m1}, \mathbf{U}_{m2}) \leq u_{m2} - a_{m2} \leq 0, \quad S_L(\mathbf{U}_{m2}, \mathbf{U}_{m3}) \leq u_{m2} - a_{m2} \leq 0,$$

the interface of $(\mathbf{U}_{m1}, \mathbf{U}_{m2})$ is transonic (the first subsonic interface from left to right).

Comparing with Theorem 1, the interface of $(\mathbf{U}_{m1}, \mathbf{U}_{m2})$ in Davis's algorithm is similar to that of $(\mathbf{U}_m, \mathbf{U}_R)$ in Einfeldt et al.'s algorithm. The carbuncle instabilities might occur if the density jump on this interface is large enough. In order to better understand this problem, we introduce a concept of critical density.

Each discrete distribution $\mathbf{U}_{m1}, \mathbf{U}_{m2}, \dots$ may correspond to a viscous profile, which is regarded as a solution of a system of truncation error equations (modified equations). One-dimensional viscous shock profile (vector) of conservative quantity is denoted as $\mathbf{T}(x)$ and satisfies the following equations [22]

$$\frac{\partial}{\partial x} \mathbf{F}(\mathbf{T}) = \frac{1}{2} \Delta x \frac{\partial}{\partial x} \left(|\mathbf{A}(\mathbf{T})| \frac{\partial \mathbf{T}}{\partial x} \right), \quad \lim_{x \rightarrow -\infty} \mathbf{T}(x) = \mathbf{U}_L, \quad \lim_{x \rightarrow +\infty} \mathbf{T}(x) = \mathbf{U}_R,$$

where upwind dissipation matrices $|\mathbf{A}(\mathbf{U})| = |\mathbf{F}'(\mathbf{U})|$.

According to steady shock properties, the left eigenvalue $\lambda_1(\mathbf{T}(x))$ (corresponding to S_L) must be zero somewhere along a trajectory connecting $(\mathbf{U}_L, \mathbf{U}_R)$. In fact, the conclusion comes from the continuity of viscosity profile,

$$\lambda_1(\mathbf{T}(-\infty)) = \lambda_1(\mathbf{U}_L) = u_L - a_L > 0 > u_R - a_R = \lambda_1(\mathbf{U}_R) = \lambda_1(\mathbf{T}(\infty)). \quad (33)$$

We denote the critical state as $\mathbf{U}_{cri} = \mathbf{T}(x_{cri})$, which x_{cri} is the critical transonic point from supersonic state to subsonic state, $\lambda_1(\mathbf{U}_{cri}) = 0$. Especially, the critical density in $\mathbf{U}_{cri}$ is denoted as ρ_{cri} .

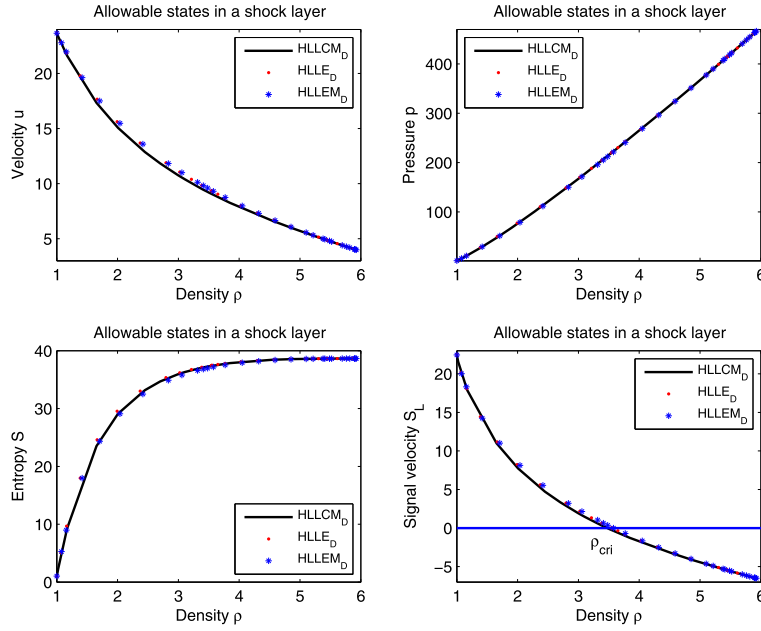


Fig. 5. The quantities in the shock layer as functions of the density for different flux functions. ρ_{cri} is a critical density, $M_a = 20$.

Table 4

The critical density ρ_{cri} of 1D stationary shock for Davis's algorithm and initial parameter.

Mach	10	20	40	...	100	α_ρ
$HLLC_D$ and $HLLCM_D$	3.37	3.48	3.50	...	3.50	0.619
$HLLM_D$ and $HLLC_D$	3.46	3.57	3.60	...	3.60	0.649
$HLLM_D$	3.44	3.55	3.57	...	3.57	0.649

Proposition 2. About density distributions on cells $m1$ and $m2$, there exist upper bounds ρ_{cri} and ρ_{cri1} such that

$$\rho_L \leq \rho_{m1} \leq \rho_{cri}, \quad \rho_{cri} \leq \rho_{m2} \leq \rho_{cri1} < \rho_R.$$

Furthermore, we have

$$\rho_{m2} \rightarrow \begin{cases} \rho_{cri}, & \text{as } \rho_{m1} \rightarrow \rho_L, \\ \rho_{cri1}, & \text{as } \rho_{m1} \rightarrow \rho_{cri}. \end{cases} \quad (34)$$

In fact, Theorem 3 tells us that the discrete distribution always has a supersonic transition state $\mathbf{U}_{m1}$ and a subsonic state $\mathbf{U}_{m2}$. If $\mathbf{U}_{m1}$ approaches $\mathbf{U}_L$, but $\mathbf{U}_{m2}$ does not converge to $\mathbf{U}_{cri}$, then the discrete shock profile will transit from $\mathbf{U}_L$ to a subsonic state $\mathbf{U}_{m2}$ directly. This is contrary to Theorem 3. Thus at this time, $\mathbf{U}_{m2}$ goes to $\mathbf{U}_{cri}$.

One can obtain critical state $\mathbf{U}_{cri}$ by numerical calculation. By varying the initial value parameter α_ρ from 0 to 1, one may reach any value $\rho_{m1} \in (\rho_L, \rho_{cri}]$. Let $\rho_{m1} \rightarrow \rho_L$, then ρ_{m2} is a good approximation to ρ_{cri} .

Plotting all discrete states in a phase space, we have a graphical representation to describe the relation of physical variables. Fig. 5 illustrates velocity, pressure, entropy and left signal velocity in a shock layer as functions of density for three different numerical schemes, in which upstream Mach number is $M_a = 20$.

From Fig. 5, we may see that these phase curves are very close for different numerical schemes. The physical quantities are always monotonously changing with the increase of density. However, critical densities differ among different schemes. Table 4 summaries critical densities for different shock strengths, in which the initial parameters α_ρ are also given corresponding to critical states for $M_a = 100$.

Fig. 6 displays the density ρ_{m2} and density ratio ρ_{m2}/ρ_{m1} as ρ_{m1} varies from ρ_L to ρ_{cri} for three schemes. The density ρ_{m2} is a monotonic increasing function of ρ_{m1} , but the density ratio ρ_{m2}/ρ_{m1} decreases monotonously.

Proposition 3. ρ_{cri}/ρ_L is the largest density ratio between two adjacent cells in computational domain.

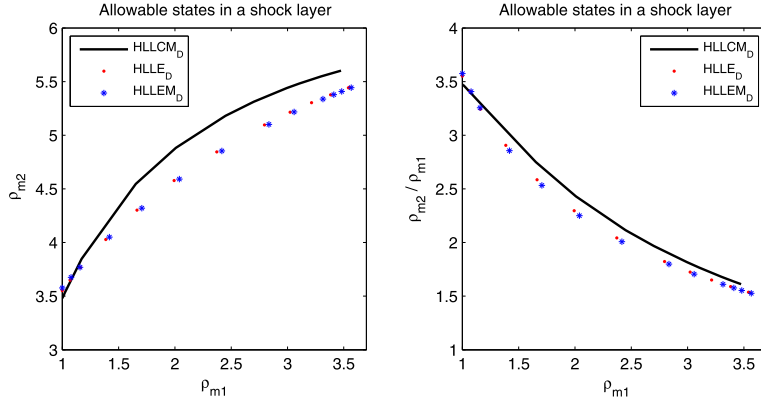


Fig. 6. The density ρ_{m2} and density ratio ρ_{m2}/ρ_{m1} as functions of the density ρ_{m1} for different flux functions, $M_a = 20$. $\rho_{m1} \in (\rho_L, \rho_{cri}]$. The minimum value of ρ_{m2} is ρ_{cri} , and the maximum value of ρ_{m2}/ρ_{m1} is ρ_{cri}/ρ_{m1} .

In fact, according to density relation (31), $\rho_{m2} \leq \rho_{m3} \leq \dots \leq \rho_R$, density ratios on post-shock adjacent cells satisfy

$$\rho_{m,k+1}/\rho_{m,k} < \rho_R/\rho_{m2}, \quad k = 2, 3, \dots$$

Notice that ρ_{cri} is the lower limit of ρ_{m2} , there is $\rho_{cri} \leq \rho_{m2}$. From Table 4, we may observe that

$$\rho_{cri}/\rho_L > \rho_R/\rho_{cri} \geq \rho_R/\rho_{m2}.$$

Thus Proposition 3 holds.

5.2. Stability analysis of Davis's algorithm

When the resulting 1D steady viscosity field is projected onto the 2D domain, the stability analysis in Section 4 can be performed. The results are very similar to those in the HLLD-type schemes. The interface of $(m1, m2)$ is the source to trigger shock instability. There exists a threshold C_{thr} so that when $\rho_{m2}/\rho_{m1} > C_{thr}$, the calculation is unstable and the most precarious case occurs as $\mathbf{U}_{m1}$ approaches $\mathbf{U}_L$.

Instead of seeking bounds of stability region of $HLLD_D$ -type schemes, we assume the thresholds are the same as those in the HLLD-type schemes. Therefore, we can judge the stability of a calculation by the following inequalities,

$$\text{if } \begin{cases} C_{thr,3} < \rho_{m2}/\rho_{m1}, & \text{a calculation with all } HLLD_D\text{-type schemes is unstable,} \\ C_{thr,2} < \rho_{m2}/\rho_{m1} < C_{thr,3}, & \text{a calculation only with the } HLLD \text{ scheme is unconditional stable,} \\ C_{thr,1} < \rho_{m2}/\rho_{m1} < C_{thr,2}, & \text{a calculation with the } HLLD, HLLD_D, HLLCM_D \text{ schemes is stable,} \\ \rho_{m2}/\rho_{m1} < C_{thr,1}, & \text{a calculation with any } HLLD_D\text{-type scheme is stable.} \end{cases} \quad (35)$$

According to Proposition 3, a more simple criterion to determine stability of the $HLLD_D$ -type schemes is

$$\text{if } \begin{cases} C_{thr,3} < \rho_{cri}/\rho_L, & \text{all } HLLD_D\text{-type schemes have danger of instability,} \\ C_{thr,2} < \rho_{cri}/\rho_L < C_{thr,3}, & \text{only the } HLLD \text{ scheme is unconditional stable,} \\ C_{thr,1} < \rho_{cri}/\rho_L < C_{thr,2}, & \text{the } HLLD, HLLD_D \text{ and } HLLCM_D \text{ schemes are stable,} \\ \rho_{cri}/\rho_L < C_{thr,1}, & \text{all } HLLD_D\text{-type schemes are stable.} \end{cases} \quad (36)$$

The HLLD-type schemes with Davis' algorithm reduce density transition magnitude between the specific interface than the corresponding schemes with Einfeldt et al.'s algorithm. Therefore they have greater stable possibility than the latters. For instance, some classical 2D examples, such as hypersonic flow over a cylinder [18], Double Mach reflection and Shock diffraction [11], do not illustrate carbuncle anomalies when using the $HLLCM_D$ flux function [24]. Further discussion is given in section 6.

Comparing critical density ρ_{cri} in Table 4 with threshold C_{thr} in Remark 3, we conclude that

Proposition 4. The $HLLD_D$ Riemann solver is stable for shock waves with arbitrary intensity, but other $HLLD_D$ -type Riemann solvers still contain instability mechanism for strong shocks.

Proposition 4 replies the question in subsection 3.1, the HLLCM and HLLD fluxes with Davis's estimates are not always stable. This is in contrast to what we observed in Fig. 2. In fact, Fig. 2 displays a special case with initial parameter $\alpha_\rho = 1$,

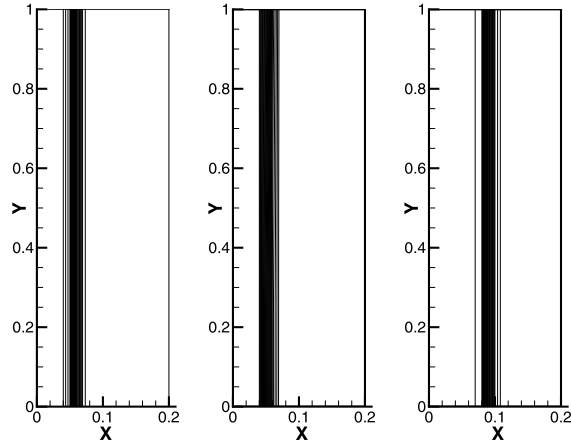


Fig. 7. The density contours for 2D steady flow with Davis' wavespeed estimates. The grid resolution is 20×100 cells. Thirty-one equally spaced contour lines from $\rho = 1$ to $\rho = 6$. From left to right: $HLLC_D$, $HLLC_M_D$, $HLLC_D$, $\alpha_\rho = 0.619$.

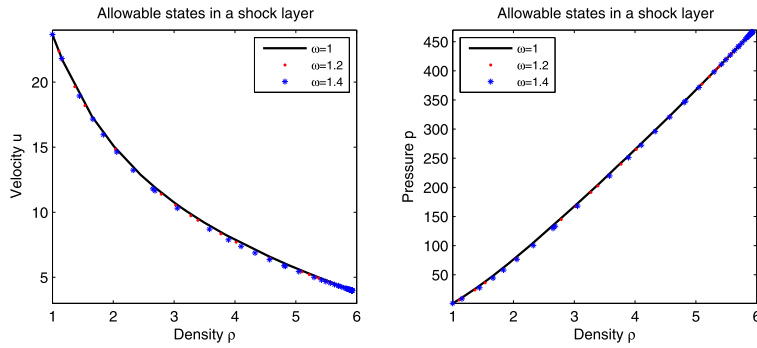


Fig. 8. The quantities in the shock layer as functions of the density for flux functions $HLLC_M_\omega$. $M_a = 20$.

which just belongs to stability domain for Davis's algorithm. If one chooses initial parameter $\alpha_\rho = 0.619$, instabilities will occur soon for the $HLLC_M_D$ and $HLLC_D$ fluxes. Fig. 7 displays the density contours with $\alpha_\rho = 0.619$.

Since the high dissipative scheme $HLLC_D$ is unconditionally stable, a hybrid formulation combining the $HLLC_D$ flux and other low dissipative flux can suppress carbuncle defeats successfully. But we prefer to use a single flux function with contact-resolving property.

5.3. Cures to the $HLLC_M$ flux and $HLLC_D$ flux

Define $HLLC_\omega$ scheme as an HLLC-type flux with the following numerical signal speeds:

$$S_L = \min(u_L - \omega a_L, u_R - \omega a_R), \quad S_R = \max(u_L + \omega a_L, u_R + \omega a_R), \quad \omega \geq 1. \quad (37)$$

If $\omega = 1$, the wave speed estimates degenerate to Davis's algorithm. If $\omega > 1$, the dissipation in an $HLLC_\omega$ scheme becomes larger than the corresponding $HLLC_D$ -type one and more transition cells appear in the shock wave layer. Fig. 8 illustrates $\rho - u$ and $\rho - p$ relations in the shock layer with different ω . The solution behaviors are almost same with $\omega = 1$.

Using the same argument as in subsection 5.1, the properties of the $HLLC_D$ -type Riemann solvers can be transplanted here. The interface of $(\mathbf{U}_{m1}, \mathbf{U}_{m2})$ is the most key location to trigger instability. The critical density ρ_{cri}/ρ_L represents the largest density ratio between two adjacent cells in computational domain. Fig. 9 displays left signal velocity S_L with different ω as a function of density. We may observe that the critical density ρ_{cri} is decreased monotonously as ω increases. This means the schemes will have better stability along with the increase of ω . Table 5 gives some critical densities for different ω , ρ_{cri} in the $HLLC_M_\omega$ flux is always smaller than that in the $HLLC_D$ flux for same ω .

Once more, we assume that the threshold C_{thr} in Remark 1 is a criterion to carbuncle instability. By comparing critical density ρ_{cri} with C_{thr} (refer to Table 5), we can get the following conclusions: the calculation is unconditional stable for the flux $HLLC_M_\omega$ when $\omega = 1.4$, for the flux $HLLC_D$ when $\omega = 1.6$.

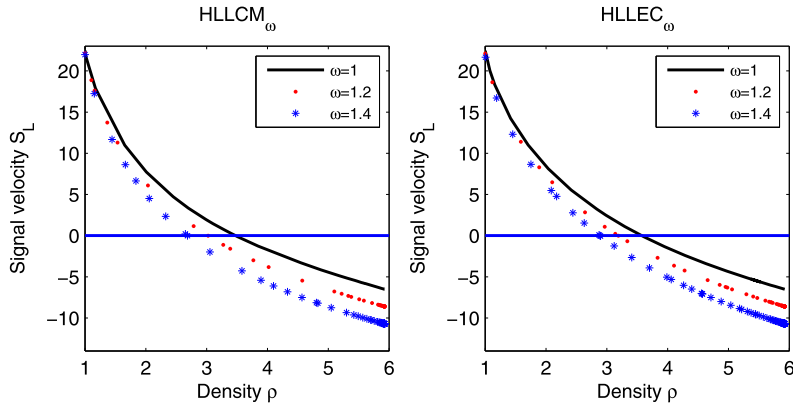


Fig. 9. The left signal velocity S_L in the shock layer as functions of the density for flux functions $HLLCM_\omega$ and $HLLC_\omega$. $M_a = 20$.

Table 5

The critical density of 1D stationary shock for the schemes of $HLLCM_\omega$ and $HLLC_\omega$, $M_a = 100$.

Fluxes	ρ_{cri}				$C_{thr,2}$
	$\omega = 1$	$\omega = 1.2$	$\omega = 1.4$	$\omega = 1.6$	
$HLLCM_\omega$	3.51	3.03	2.68	2.46	2.78
$HLLC_\omega$	3.60	3.22	2.92	2.68	2.78

Table 6

Density jump threshold ρ_R/ρ_m of instability for Einfeldt et al.'s algorithm.

Mach	$\delta = 1$		$\delta = 0.5$		$\delta = 0.1$	
	HLLCM	HLLE	HLLCM	HLLE	HLLCM	HLLE
	$C_{thr,2}$	$C_{thr,3}$	$C_{thr,2}$	$C_{thr,3}$	$C_{thr,2}$	$C_{thr,3}$
10	3.060	4.374	2.858	3.638	2.310	2.519
20	3.061	4.374	2.854	3.634	2.306	2.515
30	3.062	4.379	2.853	3.633	2.306	2.514
...						
100	3.061	4.378	2.853	3.632	2.305	2.513

6. Interpretation of the stability in other cases

By now all the matrix stability analysis for different type schemes are performed on quadrilateral uniform grids. One question is that whether the above conclusions are still valid for more general grid. We try to reply this problem by two concrete examples.

6.1. Influence of grid aspect ratio

Consider the steady shock problem (22) again. The mesh size in x-direction remains constant $\Delta x = 1$ and we change mesh size in y-direction Δy now, and the number of grid cells in both directions remains unchanged. Define an aspect ratio factor $\delta = \Delta y/\Delta x$ and compute the eigenvalues of S in function of δ and some Mach numbers. Table 6 gives theoretical results using density jump threshold C_{thr} . For brevity, only the HLLCM scheme and HLLE scheme are illustrated.

It is observed that stability threshold C_{thr} decreases with the reduction of δ . That means the carbuncle phenomenon is more severe when the grid becomes more elongated in x-direction.

When using a more dissipative wave-speed algorithm, according to the criteria (36), the stability depends on competition between C_{thr} and ρ_{cri}/ρ_L , the latter is the critical density ratio in the viscosity shock layer. Note that

$$C_{thr,2}(\delta = 1) > C_{thr,2}(\delta = 0.5) > \rho_{cri}/\rho_L(\omega = 1.4) = 2.68 > C_{thr,2}(\delta = 0.1), \text{ for HLLCM,}$$

$$C_{thr,3}(\delta = 1) > C_{thr,3}(\delta = 0.5) > \rho_{cri}/\rho_L(\omega = 1.0) = 3.57 > C_{thr,3}(\delta = 0.1), \text{ for HLLE.}$$

Thus the two shock-stable schemes on a square grid will become no longer stable if aspect ratio of grid continuously changes with reducing Δy . This fact indicates that it is impossible to construct a universal and practical shock-stable scheme

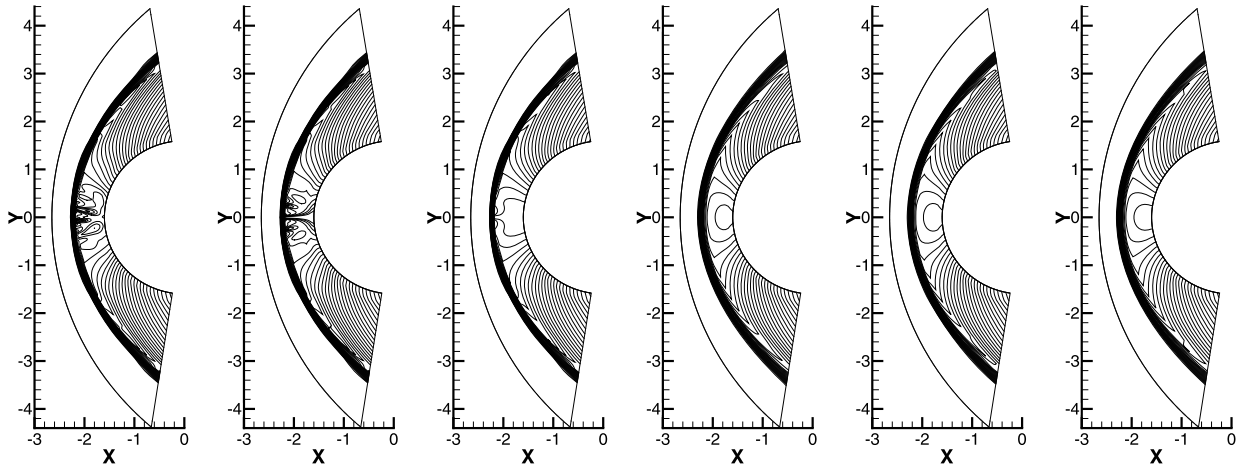


Fig. 10. The density contours for a Mach 20 blunt body with different wave velocity estimates. The grid resolution is 20×400 cells. Thirty equally spaced contour lines from $\rho = 1.5$ to $\rho = 6.5$. The first three schemes from left to right are *HLLC*, *HLLCM* and *HLLC* with Einfeldt et al.'s algorithm and the last three are *HLLC_D*, *HLLCM_D* and *HLLC_D* with Davis's algorithm.

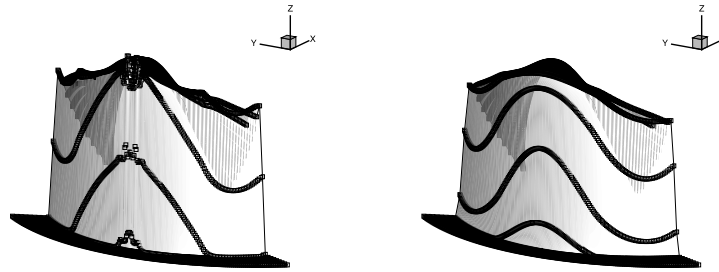


Fig. 11. The density for a Mach 20 blunt body with different wave velocity estimates. The left plot is *HLLCM* scheme with Einfeldt et al.'s wavespeed algorithm and the right is *HLLCM_D* scheme with Davis's wavespeed algorithm.

to apply for all grids. However, once grid aspect ratio is determined, shock-stable HLLC-type scheme can still be designed by varying internal structure of viscosity shock.

6.2. Unstable mechanism in a blunt body problem

In this subsection, we describe solution behaviors of the HLLC-type solvers in a blunt body problem. Fig. 10 demonstrates some numerical results with different wave-speed estimates, in which three schemes with shear viscosity are selected: the HLLC, HLLCM and HLLC schemes. For those schemes with Einfeldt et al.'s wavespeed algorithm, all the shock surfaces are preserved although some noticeable disturbances appear. Davis's wave velocity estimates are enough to suppress carbuncle phenomena. In short, stability of a blunt body calculation is better than corresponding planar shock wave.

Let's explain why the calculation of a blunt body problem has better stability than a planar shock wave with same mesh number and upstream Mach number. In fact, the generating mechanism of carbuncle solution is the same in these two test problems: a large density jump on shock front. But some differences in solution behavior still exist between the two problems. For a detached shock ahead of the blunt body, the shock strengths decrease along shock surface deviating from the stagnation line, and the largest density jumps appear at the leading edge of the bow shock due to normal shock, see Fig. 11. Thus carbuncle solution, if it happens, focus on a few meshes in the vicinity of the stagnation line rather than whole shock surface. This is in contrast to a simple planar shock wave. For the latter, the shock strengths are same along y-direction. Instability may appear on any mesh cell along shock surface. Therefore instability of a planar steady shock is more likely to appear than a blunt body shock. If a stability analysis of the steady shock is performed on the blunt body grid, then stability threshold C_{tr} of the bow shock should be bigger than C_{tr} of corresponding planar shock.

When using a more dissipative wavespeed algorithm in the blunt body problem, normal viscosity in internal shock structure is introduced. At this time, the stability of a calculation is determined by the competition between C_{thr} and ρ_{cri}/ρ_L . For these three 'shock-stable' schemes, due to existence of shear viscosity, C_{tr} might be always greater than ρ_{cri}/ρ_L , and thus stability of these schemes rely chiefly on internal shock structure. The numerical performance is that wave speed choice has more significant effect to stability than the choice of numerical scheme, as illustrated in Fig. 10.

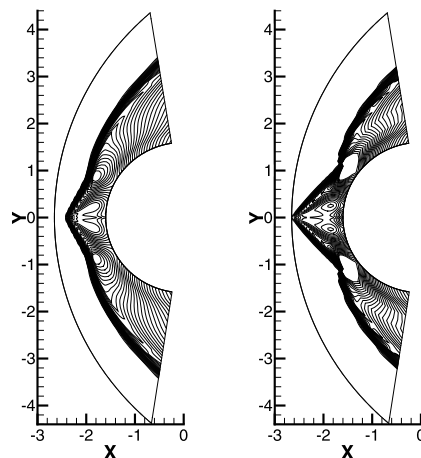


Fig. 12. The density contours for a Mach 20 blunt body with Davis's wave velocity estimates. The grid resolution is 20×400 cells. Left: $HLLC_D$, right: $HLLEM_D$. Without shear viscosity, both the calculations are unstable.

For those schemes without shear viscosity, such as the HLLC scheme or HLLEM scheme, C_{tr} might become so small that $C_{tr} < \rho_{cri}/\rho_L$, then normal viscosity from the wavespeed algorithm is not enough to suppress carbuncle phenomenon. Concrete example is that $HLLC_D$ and $HLLEM_D$ schemes still produce classical spurious protrusion along the stagnation line, see Fig. 12.

7. Conclusions

A linear stability analysis is reviewed for the HLLC-type schemes to seek the source to produce carbuncle phenomena. Here the HLLC-type schemes are with Einfeldt et al.'s wave speed algorithm and thus can accurately resolve single stationary shock wave. For a given scheme, there exists a threshold that instability occurs when density ratio between a specific interface exceeds this threshold. The specific interface is the transonic one and is the source of instability.

In order to cure instability of the HLLC-type schemes, Davis's wavespeed algorithm is discussed. Same as the cases with Einfeldt et al.'s estimates, the transonic interface is also the most key location to trigger instability. Since Davis's wavespeed estimations introduce more dissipations and reduces density jump in the viscous shock layer, they provide better stability than the corresponding Einfeldt et al.'s schemes. On a square grid, the HLLC Riemann solver with Davis's algorithm is proved to be unconditional stable for shocks with arbitrary intensity.

Introducing more dissipations in a 1D flux formulation is a way to cure carbuncle instability for other HLLC-type schemes. This paper designs unconditional stable HLLCM and HLLC schemes by relaxing signal velocities on a square grid.

It needs to point out that the stabilizing role by adjusting signal velocities of the HLLC-type schemes is limited. Carbuncle phenomena perhaps appear on more general grids even if 'shock-stable' scheme is used. A universal numerical method applying for all aspect ratio grids seems impossible. However, the theoretical results in this paper can still provide help to better understand carbuncle phenomena in general shock problems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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